

Product Analyst Interview Prep

Electronic Arts · The Sims Franchise · Hyderabad

Built for Irfan Ansari | Prepared from the job description, your resume, and your last-round debrief

10-Day Study Plan + Question Bank + Statistical Reasoning Drills

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How This Guide Is Built

This guide is built directly from the EA Product Analyst (The Sims) job description mapped against your resume, plus the specific question you got tripped up on in your last interview. Four things stand out when the two are lined up side by side:

- **Strong overlap:** your SQL, dashboarding (Metabase/Power BI), A/B testing, and KPI-definition experience at Playo map almost one-to-one onto the "What You'll Do" list. Lead with these.
- **Vocabulary gap, not skill gap:** the JD uses gaming/live-service language — "telemetry," "game health," "live operations," "player behavior" — where your resume says "platform health," "engagement," "user behavior." Same muscles, different words. Practice translating your Playo stories into this vocabulary.
- **Thin spot:** "statistical modeling / machine learning techniques" and "determining statistical significance" are called out explicitly. Your resume doesn't foreground statistics theory, so this guide weights that section heavily.
- **The gap that already cost you a beat:** a conceptual statistics question (does $A > B$ in every subgroup imply $A > B$ overall?) is exactly the kind of "reasoning under the hood" question analytics interviewers use to test whether you understand your numbers or just run queries. Section 5 drills this pattern hard, with your hospital example fully worked out.

How to use it: Don't try to memorize answers. For every question, say it out loud in your own words, using a real example from Playo (Users Matrix Dashboard, Corporate Coupon A/B test, Weekly Business Dashboard, Year in Playo). Interviewers remember specific numbers and decisions, not definitions.

10-Day Study Plan

Day	Focus	What to actually do
Day 1	SQL fundamentals refresh	Redo joins, window functions (RANK, LAG/LEAD, ROW_NUMBER), CTEs, and cohort-style self-joins. Rewrite 3 of your real Playo queries from memory without looking at the original.
Day 2	SQL for product metrics	Practice writing retention (D1/D7/D30), funnel conversion, and DAU/MAU queries from raw event tables. Time yourself — aim for a working query in under 10 minutes.
Day 3	Statistics & probability core	Go through Section 4 of this guide. Focus on p-values, confidence intervals, Type I/II errors, and distributions. Explain each concept to yourself with a Playo example.
Day 4	Statistical reasoning & paradoxes	Work through every question in Section 5, including the hospital example, with pen and paper. Redo the arithmetic from scratch until it's automatic.
Day 5	A/B testing design	Section 2. Practice designing a full test end-to-end out loud: hypothesis → metric → sample size → guardrails → analysis → decision. Use your Corporate Coupon project as the running example.
Day 6	Gaming / live-service product sense	Section 1. Read up on live-service and game-health concepts (DAU/MAU stickiness, session length, ARPPAU, churn, LTV, expansion-pack launches). Practice 5 case questions out loud.
Day 7	The Sims & EA context	Play or watch gameplay of a recent Sims expansion; skim EA's investor/live-services commentary for The Sims. Note 3 things you'd want to measure if you joined that team tomorrow.
Day 8	Full mock interview (technical)	Simulate 60 minutes: 2 SQL questions, 1 stats/paradox question, 1 A/B design question, 1 case question. Say answers out loud or record yourself.
Day 9	Behavioral + storytelling	Section 6. Turn your 5 Playo/NoBroker projects into tight STAR stories (Situation, Task, Action, Result) with numbers. Prepare 3 questions to ask the interviewer.
Day 10	Light review + rest	Skim the cheat sheet (Section 7) only. No new material. Sleep well — reasoning questions are where tired candidates slip.

Section 1 · Product & Case Questions (Game Health, Live Ops, Telemetry)

These map to: "analyze and monitor game health metrics," "explore player behavior," "define telemetry, success metrics, KPIs," and "translate ambiguous business questions into structured analyses."

Q1. How would you define "game health" for The Sims, and what metrics would you track weekly vs. daily?

APPROACH

- Separate into layers: *acquisition* (installs, activations), *engagement* (DAU/MAU, session length, session frequency, stickiness = DAU/MAU), *retention* (D1/D7/D30, rolling retention curves), *monetization* (ARPPAU, conversion to paying, pack/DLC attach rate), and *stability/ops* (crash rate, load times, error rates).
- Daily: crash rate, DAU, server/latency health, live-event participation. Weekly/monthly: retention cohorts, monetization trends, feature adoption — daily noise washes out and seasonality (weekday vs weekend play) becomes visible.
- Tie back to your Weekly Business Dashboard and Users Matrix Dashboard projects as a direct parallel.

Q2. DAU is flat but MAU is growing. What's happening and what would you check?

APPROACH

- Stickiness (DAU/MAU) is falling — you're acquiring/reactivating users but not deepening engagement per user.
- Segment by cohort age (new vs. returning), platform, and country before concluding anything.
- Check for confounds: a marketing push inflating MAU with low-intent users, a content drought reducing daily habit, or a technical issue (notifications broken, login bug) suppressing daily visits for a subset.
- Mention you'd cross-check with session length and D7 retention of new cohorts specifically — this is the "case-mix" instinct from Section 5.

Q3. A new Sims expansion pack launches. What's your measurement plan for the first 30 days?

APPROACH

- Pre-launch: define success metrics with PM/design before launch (adoption %, D1/D7 retention of buyers vs. non-buyers, attach rate, revenue, sentiment/ratings, support ticket volume, crash rate by device).
- Launch day/week: monitor real-time telemetry for crashes, server load, and early funnel drop-off (store view → purchase → first play).
- 30-day view: cohort retention of pack buyers vs. base game players, cannibalization check (did existing content usage drop), and halo effect on overall game DAU.
- Close with: "I'd present this as a dashboard the whole live-ops team can self-serve, the way I built the Weekly Business Dashboard at Playco."

Q4. How would you design telemetry/instrumentation for a brand-new gameplay feature that doesn't exist yet?

APPROACH

- Start from the question, not the event: "what decision will this data inform?" then work backward to the minimum events needed.
- Define event schema: event name, timestamp, user_id, session_id, feature-specific parameters (e.g., object_placed, room_type, interaction_duration).
- Plan for both usage metrics (adoption, frequency) and outcome metrics (does it change retention/monetization/session length).
- Flag data quality guardrails up front: required fields, validation rules, deduplication logic — this is literally your "data validation and quality checks" bullet from Playco.

Q5. Engagement dropped 15% after a patch. Walk me through your investigation.

APPROACH

- Confirm it's real first: check instrumentation/logging didn't break with the patch (a very common false alarm).
- Segment: by platform (console/PC/mobile), region, new vs. existing users, patch-adopters vs. not-yet-updated.
- Timeline: is it a step-change at patch release or a gradual decline (rules out vs. supports the patch as cause).
- Correlate with patch notes: removed content, new bugs, changed difficulty/economy balance, or store price changes.
- Conclude with a recommendation and how you'd validate it (e.g., propose a fix, monitor for reversal, or A/B test a rollback of one change).

Q6. How do you decide which metric is the "North Star" for a live-service game vs. vanity metrics?

APPROACH

- North Star should reflect sustained value delivered to players and predict long-term business health (e.g., engaged retained players, not raw downloads).
- Vanity metrics move easily (installs from a paid campaign) but don't correlate with LTV or retention.
- Good North Star for The Sims-type game: something like "weekly active engaged players" (session length/frequency threshold) or a composite retention+engagement score, supported by a metric tree of inputs underneath it.

Q7. How would you identify an "underperforming segment" of players, and what would you do next?

APPROACH

- Define segments meaningfully first (new vs veteran, platform, spend tier, geography) rather than slicing arbitrarily until something looks bad (multiple-comparisons trap — mention this explicitly, it signals rigor).
- Compare against a clear benchmark (cohort average, prior period, similar segment).
- Root-cause with a mix of quantitative (funnel, telemetry) and qualitative (support tickets, reviews) evidence before recommending action.

Q8. How would you build a self-serve dashboard for a live-ops team that isn't technical?

APPROACH

- Start with 3-5 questions the team actually asks weekly, not every metric you can compute.
- Design for the decision, not the data: clear thresholds/alerts, drill-down from summary to detail, consistent definitions documented alongside the dashboard.
- Reuse your real example: Metabase dashboards at Playo used by Product/Marketing/Ops without needing you to pull ad hoc numbers each time.

Q9. Two features both look "successful" in isolation, but the team can only fund one going forward. How do you help decide?

APPROACH

- Normalize both to the same unit of value (incremental retention days, incremental revenue per user) rather than comparing raw adoption %.
- Account for cost/effort to maintain and opportunity cost, not just measured lift.
- Check for overlap/cannibalization — are they competing for the same player time or the same use case?
- Present a trade-off, not just a number: this is where "translate ambiguous questions into actionable recommendations" from the JD shows up directly.

Q10. How would you keep analytics pipelines reliable as The Sims scales telemetry across platforms (PC, console, mobile)?

APPROACH

- Standardize event schemas across platforms early; a shared event dictionary avoids "three definitions of session_start."
- Automated data quality checks (null rates, volume anomalies, schema drift) with alerting — directly your Airflow automation experience at Playo.
- Version and document metric definitions so dashboards don't silently diverge when someone tweaks a query.

Q11. How would you explain a drop in a key metric to a non-technical stakeholder (e.g., a producer) who wants a one-line answer, when the real cause is nuanced?

APPROACH

- Lead with the headline and the confidence level, then one supporting layer of "why," and offer more detail if asked — don't front-load methodology.
- Use a concrete number and a comparison ("down 12% week-over-week, driven mostly by EU mobile users after Tuesday's patch") rather than abstractions.
- Be explicit about what's confirmed vs. hypothesis, and what you're doing next.

Q12. What would you want to know in your first 30 days on The Sims analytics team?

APPROACH

- Existing metric definitions and known data quality issues (avoid re-deriving DAU a different way than everyone else).
- Current experimentation cadence and tooling (in-house vs. third-party A/B platform).
- Roadmap for upcoming live-ops events/expansions so you can plan instrumentation ahead of launch, not after.
- This question rewards curiosity and "player-first" thinking — mention wanting to actually play the game to build intuition for the telemetry.

Section 2 · A/B Testing & Experimentation

Q13. Walk me through how you'd design an A/B test end-to-end.

KEY POINTS

- Hypothesis (specific, falsifiable) → primary metric + guardrail metrics → unit of randomization (user vs. session vs. device) → minimum detectable effect & sample size → test duration (cover full weekly cycle) → run → check sample ratio mismatch → analyze → decide & document.
- Use your Corporate Coupon Analysis (new vs. old users, before/after) as your real, talkable example — be ready to explain why it was a quasi-experiment, not a true randomized A/B test, if pressed.

Q14. How do you calculate the sample size / duration needed for a test?

KEY POINTS

- Depends on baseline conversion rate, minimum detectable effect (MDE), desired significance level (α , usually 0.05) and power ($1-\beta$, usually 0.80).
- Smaller MDE or lower baseline variance → larger sample needed. Formula intuition: $n \propto (z_{\alpha/2} + z_{\beta})^2 \times \text{variance} / \text{MDE}^2$.
- Always round test duration up to full weeks to capture weekday/weekend behavior cycles.

Q15. Your test shows a statistically significant result after 2 days instead of the planned 2 weeks. Do you call it?

KEY POINTS

- No — this is "peeking" / optional stopping, which inflates false positive rate well above 5%.
- Explain: with repeated peeking, you're running many implicit hypothesis tests, so the real Type I error rate compounds.
- Correct approach: pre-register duration/sample size, or use a sequential testing method designed for early stopping (e.g., alpha-spending, Bayesian approaches) if early looks are a business requirement.

Q16. How do you pick a primary metric vs. guardrail metrics for an experiment?

KEY POINTS

- Primary metric = the thing the feature is meant to move, ideally close to business value (e.g., D7 retention, revenue per user) not just a proxy (clicks).
- Guardrails = metrics that must not regress even if the primary improves (crash rate, load time, unsubscribe rate, other feature usage) — protects against "winning" a metric while quietly harming the product.

Q17. What's the difference between statistical significance and practical significance?

KEY POINTS

- Statistical significance = unlikely the observed difference is due to chance, given the sample (p-value below threshold).
- Practical significance = the effect is large enough to matter for the business/player experience.
- With huge sample sizes (common in live-service games), tiny, meaningless differences can become "statistically significant" — always report effect size and confidence interval, not just a p-value.

Q18. You're testing 6 metrics in one experiment and one comes back significant. Do you ship it?

KEY POINTS

- Multiple comparisons problem — testing 6 metrics at $\alpha=0.05$ gives a much higher than 5% chance that at least one looks significant by chance alone.
- Mitigate with a pre-specified primary metric, correction methods (Bonferroni, Benjamini-Hochberg/FDR) for secondary metrics, or treat unplanned significant secondaries as hypotheses to re-test, not conclusions.

Q19. How would you A/B test something you can't randomize at the individual user level (e.g., a server-region change or a pricing change with network effects)?

KEY POINTS

- Consider switchback tests (alternate treatment over time) or geo/cluster-based randomization (randomize by region/server shard instead of user).
- For network effects or pricing, consider quasi-experimental designs: difference-in-differences, or a holdout region compared pre/post — and be explicit about the weaker causal claim versus a true RCT.

Q20. An experiment shows a positive lift overall, but your PM asks "are you sure it's not just Simpson's Paradox from an uneven rollout?" How do you check?

KEY POINTS

- This is the A/B-testing version of Section 5's core idea. If traffic allocation to treatment/control shifted over time or across segments (e.g., ramp-up rollout, or mobile vs. console mix changed), the pooled result can reverse or mask subgroup effects.
- Check: (1) sample ratio mismatch over time, (2) consistent effect direction within each major segment (platform, region, new vs. existing users), (3) re-run the analysis with time period or segment as a stratifying/control variable.
- This is a great moment to explicitly say "Simpson's Paradox" out loud — ties directly back to your last interview's question.

Section 3 · SQL Practice & Theory

The JD calls out "complex SQL queries and scripting workflows" over "large-scale telemetry datasets." Expect both concept questions and a live query exercise.

Q21. Write a query to compute D7 retention: the % of users active on day 7 after their install date.

```
SELECT
  COUNT(DISTINCT CASE WHEN a.event_date = i.install_date + INTERVAL '7 day'
    THEN a.user_id END) * 1.0
  / COUNT(DISTINCT i.user_id) AS d7_retention
FROM installs i
LEFT JOIN activity a
  ON a.user_id = i.user_id
GROUP BY i.install_date;
```

TALKING POINTS

- Mention edge cases: users who install but never appear in `activity` at all, timezone handling for "day," and whether retention should be "on exactly day 7" vs. "day 7 or later" (rolling retention).

Q22. Find the second-highest spending player per country (classic window function question).

```
SELECT country, user_id, total_spend
FROM (
  SELECT country, user_id, total_spend,
    DENSE_RANK() OVER (PARTITION BY country ORDER BY total_spend DESC) AS rnk
  FROM player_spend
) t
WHERE rnk = 2;
```

TALKING POINTS

- Explain `RANK` vs `DENSE_RANK` vs `ROW_NUMBER` and when ties matter for "second highest."

Q23. How would you detect a data quality issue like duplicate event logging in a telemetry pipeline, using SQL?

KEY POINTS

- Group by natural key (`user_id`, `event_name`, `timestamp`, `session_id`) with `HAVING COUNT(*) > 1` to surface exact duplicates.
- Check volume anomalies by comparing daily event counts against a trailing average (z-score or % deviation) to catch silent double-firing after a client release.
- Tie to your resume: "implement data validation and quality checks across multiple sources."

Q24. Explain the difference between `WHERE` and `HAVING`, and when a window function is preferable to a self-join.

KEY POINTS

- `WHERE` filters rows before aggregation; `HAVING` filters after grouping/aggregation.
- Window functions avoid the row-multiplication and performance cost of self-joins when you need "value relative to a partition" (running totals, rank, lag/lead) while keeping row-level granularity.

Q25. You're given raw MongoDB-style nested JSON event logs and need to compute daily active users in SQL. How do you approach it?

KEY POINTS

- Flatten/normalize nested fields first (or query via a JSON-aware layer / dbt staging model) before aggregation — directly maps to your MongoDB + MySQL + Firebase experience at Playo.
- Deduplicate by `user_id` + `date` before counting distinct users; watch for multiple device IDs per user without a stitched identity.

Section 4 · Statistics & Probability Theory

Q26. Explain a p-value in plain language, the way you'd explain it to a producer.

KEY POINTS

- "If there were truly no difference between the two versions, a p-value of 0.03 means there's a 3% chance we'd see a difference this large (or larger) just from random noise."
- Common mistake to call out: a p-value is *not* "the probability the null hypothesis is true," and it's not "the probability your result is due to chance" in a Bayesian sense — it's a statement about the data given the null.

Q27. What are Type I and Type II errors, and how do they trade off?

KEY POINTS

- Type I (α): false positive — shipping a feature that doesn't actually help. Type II (β): false negative — killing a feature that actually would have helped.
- Lowering α (stricter significance bar) reduces false positives but increases false negatives for a fixed sample size; the only way to reduce both is more data (larger sample / longer test).
- Frame the business trade-off: in a live-service game, shipping a bad feature (Type I) may be more costly than missing a small good one (Type II), or vice versa depending on rollback ease.

Q28. What's a confidence interval, and how is it different from a p-value?

KEY POINTS

- A 95% CI is a range such that, if you repeated the experiment many times, 95% of such intervals would contain the true effect — it conveys both direction and magnitude of uncertainty, which a single p-value doesn't.
- Prefer reporting CIs alongside p-values in stakeholder communication because they show the plausible size of the effect, not just "significant or not."

Q29. Correlation vs. causation — give a gaming-relevant example of where this trips people up.

KEY POINTS

- "Players who reach level 10 spend more" doesn't mean leveling causes spending — more engaged/invested players do both. A confound (underlying engagement/skill) drives both variables.
- Best real fix: randomized experiment (force some players through a faster leveling path) rather than relying on observational correlation.

Q30. What's regression to the mean, and where would you see it in live-ops data?

KEY POINTS

- Extreme results (a top-decile spending week, a worst-ever churn day) tend to be followed by more average results, purely from natural variance — not because of anything you did.
- Trap: launching an intervention right after an unusually bad metric day/week and then crediting the intervention when the metric naturally reverts.

Q31. Explain the Central Limit Theorem and why it matters for A/B testing.

KEY POINTS

- The sampling distribution of a sample mean approaches normal as sample size grows, regardless of the underlying distribution of individual data (e.g., heavily skewed spend data per player).
- This is why t-tests/z-tests on metric *averages* are valid at scale even when the raw per-user data (spend, session length) is not normally distributed.

Q32. When would you use a t-test vs. a chi-square test vs. a proportions z-test?

KEY POINTS

- T-test: comparing means of a continuous metric (e.g., average session length) between two groups.
- Proportions z-test: comparing conversion/retention rates (binary outcome) between two groups — most common in feature A/B tests.
- Chi-square: testing association between two categorical variables (e.g., platform vs. churned/retained) or comparing observed vs. expected distribution across more than 2 categories.

Q33. What does it mean for data to have "high variance," and why does it matter for experiment sample size?

KEY POINTS

- High variance means individual values are spread widely around the mean (e.g., whale spenders vs. non-payers) — makes it harder to detect a true difference in means against that noise.
- Higher variance → need a larger sample size for the same statistical power, or consider variance-reduction techniques (e.g., CUPED, stratification, capping outliers).

Q34. "Hands-on with data mining/ML techniques" is in the JD. What's a simple ML/statistical model you could apply to churn prediction, and how would you validate it?

KEY POINTS

- Logistic regression or a gradient-boosted tree (e.g., XGBoost) predicting probability of churn from engagement/monetization features — start simple and interpretable before reaching for complexity.
- Validate with a held-out time-based test set (not random split, to avoid leakage from future info), and track precision/recall or AUC rather than just accuracy, since churn is usually class-imbalanced.
- You don't need deep ML expertise for this role — showing you know the right validation instincts (avoid leakage, watch for imbalance) is enough.

Section 5 · Statistical Reasoning & Paradox Questions

These are "do you understand what's under your numbers" questions. Interviewers use them because a candidate who can write a query but can't reason about confounding will ship wrong conclusions. This section fully solves the exact question you got in your last round.

Q35 — The one you were asked: "Hospital A has a higher success rate than Hospital B in every individual type of surgery. Does that mean Hospital A's overall success rate is always higher than Hospital B's?" Simpson's Paradox

SHORT ANSWER

No. This is Simpson's Paradox: a trend that holds in every subgroup can reverse when the subgroups are combined, if the subgroups have very different sizes (weights) in each hospital's overall mix.

WORKED NUMERIC EXAMPLE (USE THIS EXACT STRUCTURE OUT LOUD)

Assume two types of surgery: **Easy** and **Hard**. Hospital A happens to take on far more hard, high-risk cases (it's the referral center); Hospital B mostly does easy cases.

	Hospital A	Hospital B
Easy surgeries	95 / 100 succeed → 95%	810 / 900 succeed → 90%
Hard surgeries	540 / 900 succeed → 60%	50 / 100 succeed → 50%
Overall	635 / 1000 → 63.5%	860 / 1000 → 86%

Hospital A beats Hospital B in *both* categories individually (95%>90%, and 60%>50%) — yet Hospital B's *overall* success rate (86%) is much higher than Hospital A's (63.5%).

WHY IT HAPPENS (THE ONE-SENTENCE EXPLANATION)

Hospital A did 900 hard, riskier surgeries and only 100 easy ones; Hospital B did the reverse. The overall rate is a *weighted* average, and the weights (case mix) differ so much between the two hospitals that they dominate the result — the "hidden" or confounding variable here is **case difficulty / patient risk mix**.

HOW TO CLOSE THE ANSWER IN AN INTERVIEW

- State the answer immediately: "No, this is a case of Simpson's Paradox."
- Give the intuition in one sentence before numbers: "It happens when the two things being compared handle very different mixes of easy vs. hard cases."
- Then walk through a simple numeric example (you can use round numbers like above, you don't need to memorize this exact table — construct one live: pick two categories, make Hospital A dominate the risky category and Hospital B dominate the easy one).
- Connect it to the job: "In an analytics role, this is why I'd never trust an aggregate metric without checking segment mix first — e.g., in an A/B test, if the treatment group's traffic composition shifted over the test, the overall lift can be misleading even if every segment individually improved."

Q36. A medical test is 99% accurate for a disease that affects 1 in 10,000 people. You test positive. What's the actual probability you have the disease? Base Rate Fallacy

ANSWER

Far less than 99% — roughly **~1%**. With 10,000 people, ~1 person actually has the disease (and tests positive with 99% probability). But among the other 9,999 healthy people, a 1% false-positive rate produces about 100 false positives. So of ~101 positive tests, only 1 is a true case → $P(\text{disease} \mid \text{positive}) \approx 1/101 \approx 1\%$.

WHY IT'S ASKED

- Tests intuition for base rates / prior probability — directly relevant to interpreting any low-incidence event (fraud detection, churn flags, rare bugs) where a "high accuracy" model can still produce mostly false alarms.

Q37. You look at survey data: players who use a certain in-game item report much higher satisfaction. Should you roll it out to everyone? Survivorship / Selection Bias

ANSWER

Not on this evidence alone. Players who chose to use the item may already be more engaged, more skilled, or spend more — the item's users are a self-selected, non-random sample. You're seeing correlation between "opts into item" and "satisfaction," not proof the item causes satisfaction. Need a randomized rollout (A/B test) to isolate the causal effect.

Q38. A feature's retention lift looked great in the first week of the A/B test, then shrank to nothing by week 3. What might explain this? **Novelty Effect**

ANSWER

Classic novelty effect: users try something new because it's new, which temporarily boosts engagement metrics regardless of the feature's real long-term value. Also consider a "peeking" false positive in week 1, or a seasonal effect. Correct approach: run tests long enough to look past the novelty window, and specifically track whether the lift is sustained, not just present at day 3.

Q39. Average revenue per user went up 20% last month, but the median stayed flat. What does that tell you, and what should you check? **Mean vs. Median / Outlier Sensitivity**

ANSWER

The mean is being pulled up by a small number of very high spenders ("whales") while the typical player's spend didn't change. This is a sign of a right-skewed distribution. You should report both mean and median (and ideally percentiles/whale-vs-non-whale segments) rather than a single average, since the average alone overstates the typical player experience — important if this metric is about to justify a broad product decision.

Q40. Three doors, one has a prize. You pick one; the host (who knows what's behind each) opens a different door showing no prize, then offers you the choice to switch. Should you switch? **Monty Hall Problem**

ANSWER

Yes, switch — switching wins 2/3 of the time, staying wins 1/3. Intuition: your original pick had a 1/3 chance of being right when you chose it, and that doesn't change just because the host reveals information. The host's guaranteed reveal of a non-prize door concentrates the remaining 2/3 probability onto the one door you didn't pick.

WHY IT'S ASKED

Tests comfort with conditional probability and resisting a strong, wrong intuition — the same muscle you need when a "significant" result is actually an artifact of how the data was filtered or revealed to you.

Q41. Two independent A/B tests each show a "positive but not quite significant" result ($p = 0.08$ both times) for the same feature on two different platforms. Can you combine them to claim significance? **Multiple Testing / Meta-analysis Caution**

ANSWER

Not by simply eyeballing two "close" p-values as reinforcing each other — that's an intuitive but statistically invalid shortcut. The correct approach is a pre-planned pooled analysis (combine the raw data with platform as a covariate) or a formal meta-analytic method, not "two nearly-significant results feel like one significant result." Also worth flagging: if you ran the test on two platforms and are now combining because neither alone worked, that's an ad hoc/post hoc analysis choice, which itself should be treated cautiously.

Q42. Follow-up they may ask: "Can you think of another business example (outside hospitals) where this same trap shows up?"

HAVE 2 READY

- **Hiring/promotion rates by department:** a company can have a higher promotion rate for women than men in every single department, yet a lower overall rate for women, if women are concentrated in larger/more competitive departments with lower promotion rates department-wide (this is the real-world Berkeley admissions-style example).
- **A/B test rollout:** a feature can outperform control in every weekly cohort, but underperform overall if the mix of new vs. returning users changed week to week during the ramp-up (ties directly back to Q20).

Section 6 · Behavioral Questions (Tied to Your Resume)

Prepare each as a tight STAR story (Situation → Task → Action → Result) with a number in the result. Suggested project to use is listed for each.

Q43. "Tell me about a time you translated an ambiguous business question into a structured analysis."

USE

Weekly Business Dashboard or Corporate Coupon Analysis — describe the vague ask you started with, how you scoped it into specific metrics, and the decision it enabled.

Q44. "Describe an A/B test or before/after analysis you ran. What was the result and what did you recommend?"

USE

Corporate Coupon Analysis (new vs. old users). Be ready for the natural follow-up: "was this a true randomized A/B test or an observational before/after comparison, and what are the limitations of that?" — answer honestly; this is a great moment to show statistical maturity by naming the limitation yourself (e.g., seasonality or selection confound) before they ask.

Q45. "Tell me about a time your analysis was wrong, or a dashboard had a data quality issue. What did you do?"

USE

Your data validation work across MongoDB/MySQL/Firebase — pick one concrete bug you caught (or should have caught earlier), what you changed in your process afterward, and how you now safeguard against it.

Q46. "How have you influenced a product decision using data, even when stakeholders initially disagreed?"

USE

Any dashboard/finding at Playo that changed a roadmap or marketing decision — emphasize how you communicated it (not just the analysis) and what pushback you handled.

Q47. "Why The Sims / EA, and why gaming analytics specifically?"

PREP

- Connect your genuine interest (or willingness to develop it) in gaming with the analytical challenge of live-service data at scale — millions of concurrent telemetry events, real-time decisions, long-lived player relationships.
- Mention something specific about The Sims (its player-driven creativity, long content lifecycle via expansion packs, multi-platform base) to show you did the homework, not a generic answer.

Section 7 · Quick-Reference Cheat Sheet

Core gaming/live-service metrics

Metric	Definition
DAU / MAU	Daily / Monthly Active Users
Stickiness	$DAU \div MAU$ — how often the average monthly user shows up daily
D1 / D7 / D30 retention	% of a cohort still active exactly N days (or "N days or later," if rolling) after install/first play
ARPDau	Average Revenue Per Daily Active User
ARPU / ARPPU	Avg Revenue Per User / Per Paying User
LTV	Lifetime Value — total expected revenue from a user over their lifecycle
Churn rate	% of active users who stop being active in a given period
Attach rate	% of base-game owners who also purchase a given DLC/expansion

Statistics quick formulas

Concept	Notes
95% Confidence Interval (proportion)	$p \pm 1.96 \times \sqrt{p(1-p)/n}$
Sample size driver	$n \propto (Z_{\alpha/2} + Z_{\beta})^2 \times \text{variance} \div MDE^2$ — smaller effect or lower baseline variance needs bigger n
$p < 0.05$	Conventional threshold for "statistically significant," not a magic truth cutoff
Type I error (α)	False positive: conclude an effect exists when it doesn't
Type II error (β)	False negative: miss a real effect. Power = $1 - \beta$

Reasoning traps to always check before trusting an aggregate number

1. Has the segment mix (case mix, traffic mix, user mix) changed or differed between the two things being compared? → Simpson's Paradox risk.
2. Is the base rate of the event very low? → Base rate fallacy risk when interpreting "accuracy."
3. Was the sample self-selected rather than randomized? → Selection/survivorship bias.
4. Did you look at the result multiple times before calling it? → Peeking / multiple comparisons.
5. Is the effect driven by a few extreme values? → Check median/percentiles, not just the mean.
6. Could this just be reversion after an extreme data point? → Regression to the mean.

Day-of reminder: For every technical or reasoning question, say your answer first in one sentence, then justify it. Interviewers grading analytics candidates are listening for structured thinking under ambiguity as much as the "correct" formula — which is exactly what the JD means by "translate ambiguous business questions into structured analyses and measurable outcomes."